Primary Ovarian Insufficiency (POI)

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Microbiome Signatures, Microbiome Medicine

Abstract. POI can cause hormonal disruptions and infertility before the age of 40. Get the facts about Primary Ovarian Insufficiency (POI), its causes, symptoms, and treatment strategies including hormone therapy, IVF, and lifestyle tips to manage health and fertility.

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Interventions

Microbiome-targeted interventions (MBTIs) are emerging as promising approaches to manage primary ovarian insufficiency. Recent studies have highlighted the role of microbiome dysbiosis, imbalances in microbial communities, in the pathogenesis of POI. Interventions aimed at restoring microbial balance in the gut and vaginal microbiota are being explored to improve ovarian function and overall reproductive health.

Class	Mechanisms of Action	MBTI Status

Probiotics	These probiotics modulate the gut and vaginal microbiota, enhancing the abundance of beneficial bacteria, reducing the presence of pathogenic microorganisms, and producing short-chain fatty acids (SCFAs) that support ovarian function.[38] Studies have shown that probiotic supplementation can improve ovarian function and delay ovarian aging by modulating gut microbiota composition and activity.[39]	Experimental
Fecal Microbiota Transplantation (FMT)	FMT aims to restore a healthy microbiota composition by introducing diverse microbial communities into the recipient's gut.[40] This intervention has shown promise in improving ovarian function and delaying ovarian aging in animal models by modulating the gut microbiota.[41]	Under Investigation
Dietary Fiber & Prebiotics	Dietary fibers serve as substrates for beneficial gut bacteria, promoting the production of SCFAs like butyrate, which have been shown to regulate estrogen and progesterone production, thereby supporting ovarian function.[42]	Promising Candidate
Hormone Replacement Therapy (HRT)	While primarily aimed at hormone regulation, HRT has been observed to influence gut microbiota composition, potentially alleviating dysbiosis associated with POI.[43] HRT reverses Gut microbiome and serum metabolome alterations in POI.[44]	First-line Treatment
Metformin (MET)	MET activates the AMPK pathway, improving mitochondrial function, reducing oxidative stress, and modulating inflammation. These effects protect ovarian granulosa cells from chemotherapy-induced damage, thus preserving ovarian function.[45]	Promising
Stem Cell Therapy	hUC-MSC exosomes mediate their therapeutic effects by regulating ovarian immune functions and reducing oxidative stress and inflammation, leading to improved ovarian function and fertility.[46]	Promising Candidate

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